

Program : Diploma in Architecture	
Course Code : 5182B	Course Title: Green Building and Energy Conservation
Semester : 5	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:4, T: 0, P: 0)	Periods per semester: 60

Course Objectives:

- To know various aspects of green buildings
- To use different steps involved in measuring environmental impact assessment.
- To relate the construction of green building with prevailing energy conservation policy and regulations.
- To know and identify different green building construction materials.
- To learn different rating systems and their criteria.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Knowledge about construction materials		Construction materials	3

Course Outcomes:

On completion of the course, student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Identify various requirements for green building.	12	Understanding
CO2	Relate the construction of green building with prevailing energy conservation policy and regulations to conduct environmental impact assessment.	14	Understanding
CO3	Explain the criteria related to rating systems for assessment of Green building	18	Understanding
CO4	Supervise green building construction using green materials.	14	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3				3		
CO2	3				3		2
CO3	3				3		2
CO4	3				3		2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify various requirements for green building.		
M1.01	Define green building and identify its components	4	Understanding
M1.02	Identify site selection strategies	4	Understanding
M1.03	Familiarize with energy conservation techniques	4	Understanding
Contents: Definition of Green Building, Benefits of Green building, Components/features of Green Building. Site selection, Site selection strategies, Landscaping, building form, orientation, building envelope and fenestration, material and construction techniques, roofs, walls, fenestration and shaded finishes, advanced passive heating and cooling techniques, Energy Efficiency, Water efficiency, Material Efficiency, Indoor Air Quality. waste reduction during construction			
CO2	Relate the construction of green building with prevailing energy conservation policy and regulations to conduct environmental impact assessment.		
M2.01	Identify Energy Management Programs	2	Understanding
M2.02	Explain Environmental Impact Assessment (EIA)	4	Understanding
M2.03	Understand the types of energy resources	4	Understanding
M2.04	Understand the process to get Environmental clearance for the civil engineering projects.	4	Understanding
	Series Test – I	1	

Contents: Energy Audit - Meaning, Necessity, Procedures, Types, Energy Management Programs. Environmental Impact Assessment (EIA): Introduction, EIA regulations, Steps in environmental impact assessment process, Benefits of EIA, Limitations of EIA, Environmental clearance for the civil engineering projects. Renewable Energy Resources - Solar Energy, Wind Energy, Ocean Energy, Hydro Energy, Biomass Energy. Non-renewable Energy Resources: Coal, Petroleum, Natural Gas, Nuclear Energy, Chemical Sources of Energy, Fuel Cells, Hydrogen, Biofuels. Energy conservation: Introduction, Specific objectives, present scenario, Need of energy conservation,			
CO3	Explain the criteria related to rating systems for assessment of Green building		
M3.01	Explain LEED India Rating System	4	Applying
M3.02	Describe Indian Green Building council (IGBC) Green rating	4	Applying
M3.03	Describe Green Rating for Integrated Habitat Assessment. (GRIHA) criteria	4	Applying
M3.04	Discuss Heating Ventilation Air Conditioning (HVAC) unit in green Building	3	Applying
M3.05	Discuss the functions of Government organization working for Energy conservation and Audit (ECA)	3	Applying
Contents: LEED India Rating System and Energy Efficiency - Introduction to (LEED) criteria, - Indian Green Building council (IGBC) Green rating, - Green Rating for Integrated Habitat Assessment. (GRIHA) criteria - Heating Ventilation Air Conditioning (HVAC) unit in green Building - Functions of Government organization working for Energy conservation and Audit (ECA) - National Productivity council (NPC) - Ministry of New and Renewable Energy (MNRE) -Bureau of Energy efficiency (BEE) Continuous Internal Assessment (CIA)			
CO4	Supervise green building construction using green materials.		
M4.01	Examine Principles and planning of Green building	4	Applying
M4.02	Interpret Environmental design (ED) strategies for building construction to improve environmental quality of buildings	5	Applying
M4.03	Familiarize Green building materials and products	5	Understanding
	Series Test – II	1	

Contents:

Principles and planning of Green building - Features: Salient features of Green Building, Environmental design (ED) strategies for building construction process. Improvement in environmental quality in civil structure

Materials - Green building materials and products- Bamboo, Rice husk ash concrete, plastic bricks, Bagasse particle board, Insulated concrete forms. reuse of waste material-Plastic, rubber, Newspaper wood, Nontoxic paint, Green roofing

Text/ Reference:

T/R	Book Title/Author
R1	Kibert, C.J., Sustainable construction: Green Building design and Delivery, John Wiley Hoboken, New Jersey.
R2	Chauhan, D S Sreevasthava, S K., Non-conventional Energy Resources, New Age International Publishers, New Delhi.
R3	O.P. Gupta, Energy Technology, Khanna Publishing House, New Delhi
R4	Jagadeesh, K S, Reddy Venkatta Rama & Nanjunda Rao, K S., Alternative Building Materials and Technologies, New Age International Publishers, Delhi.
R5	Means R S, Green Building - Project Planning and Cost Estimating, John Wiley & Sons
R6	Sharma K V, Venkateshaiah P., Energy Management and Conservation, IK International.
R7	Sam Kubba., Handbook of Green Building Design and Construction, Butterworth-Heinemann.